Team Project Specification

EXSU 500: Fundamentals of Artificial Intelligence in Medicine

Fall 2026 · Department of Surgery, McGill University · 65% of the course grade

1. Overview

Working in teams of four, you will define an artificial intelligence problem in health or medicine, obtain a publicly available dataset, train and compare a series of models, report the work as a short scientific manuscript, and present it in a short recorded video. Building an interface for your model is optional and carries a bonus.

The project is worth 65% of the course grade and is delivered in two graded stages: a Midterm Report (20%) and the Final Deliverables (45%). Two optional bonuses are available: 3% for publishing the codebase and 2% for deploying the interface publicly.

Novelty is not required. Competence, honesty, and reproducibility are. A careful comparison of three well-understood models on a well-chosen dataset, correctly evaluated and clearly reported, will score higher than an ambitious project with an unclear evaluation.

Learning objectives

By completing this project you will be able to:

1. Translate an informally stated clinical need into a specific, evaluable machine learning task.
2. Select and critically describe a real dataset, including its provenance, limitations and biases.
3. Train, tune and compare multiple model families under a defensible evaluation protocol.
4. Recognise and avoid the failure modes that most often invalidate medical AI results, in particular data leakage and inappropriate metrics.
5. Communicate a model's outputs and its uncertainty to someone who did not build it.
6. Report technical work to the standard of a scientific manuscript, and release it so that others can reproduce it.

2. Timeline

Date	Milestone	What you submit	Weight
Mon, Sept 21	Teams announced	Nothing. Teams are assigned by the instructors from the Pre-Assignment skills survey	—
Fri, Oct 2	Repository created and topic selected	Repository URL, submitted on myCourses, with an initial README naming the dataset and the problem	—

Date	Milestone	What you submit	Weight
Mon, Nov 9	Midterm Report	PDF + editable source (Word or LaTeX), submitted on myCourses	20%
Thu, Dec 3	Course wrap-up	Final session: Q&A and project questions. Nothing is submitted.	—
Fri, Dec 11	Final Deliverables	Manuscript (PDF + source), 7-minute recorded presentation, repository URL, peer evaluation	45% ± 5
Fri, Dec 11	Bonus: public code release (optional)	Public repository URL, submitted on myCourses	+3%
Fri, Dec 11	Bonus: public interface (optional)	URL of the publicly accessible deployed interface	+2%

All deadlines are 23:59 Eastern Time on the date shown. Submissions are made on myCourses.

3. Teams

- Teams consist of four students and are assigned by the instructors. Self-selected teams are not permitted.
- Teams are formed from the skills survey in the Individual Pre-Assignment, and are stratified so that each team contains at least one member with demonstrated programming experience and a mix of clinical and technical backgrounds.
- If the enrolment does not allow every team to be exactly four, a small number of teams of three or five will be formed. Expectations are scaled accordingly.
- Each team declares four accountable roles when it selects its topic: data lead, modelling lead, interface lead, and writing lead. Every member is expected to contribute to the code and to the manuscript; the roles establish who is answerable for each area, not who does all of it.

Requests to change teams must be made to the instructors in writing before Friday, September 25, 2026, and will be granted only for documented reasons. Teams that develop internal difficulties should raise them early: an unaddressed problem discovered in December cannot be repaired.

4. Dataset and problem definition

4.1 Choosing a dataset

- The dataset must be publicly available and downloadable by anyone, without an institutional agreement and without an approval process that takes longer than a few days.
- A list of representative open datasets, Representative Open Datasets, is posted on myCourses. It spans imaging, physiological signals, electronic health records, multi-omics, clinical text and

population health, and marks each entry as open, registration-required or restricted. It is a starting point rather than a closed list.

- You may propose a dataset that is not on the list. Submit it with your selection; approval is at the instructors' discretion and depends on licence, size, and whether it can realistically be obtained in time.
- Datasets requiring credentialed access, ethics approval, a data use agreement, or a signed licence are not permitted. Several well-known critical-care datasets fall into this category; their approval cycles are longer than this course.
- You may not use data collected at the MUHC, at any other hospital, or any data containing identifiable personal health information, unless that data has already been released publicly for educational or research use under an open licence. A public, de-identified, openly licensed hospital dataset is acceptable on the same terms as any other entry on the approved list. Internal, unreleased or individually obtained institutional data is not, whatever its source and whatever permission you believe you have.

> *Verify that you can actually download and load the data before you select it. A dataset you cannot open in week 6 is a project you cannot finish in week 13.*

4.2 Compute

Choose a dataset you can train on with the hardware you have. A free-tier cloud notebook environment with a single GPU and a few hours of runtime is sufficient for every project in this course, provided the dataset is chosen sensibly. Standardised, downsampled biomedical image collections are a good default. Do not select a hundred-gigabyte archive and discover in November that you cannot train on it.

4.3 Defining the task

Your problem must be stated as a specific, evaluable task: what is predicted, from what inputs, for whom, and why it would matter. "Apply AI to cardiology" is not a task. "Classify 12-lead ECGs into five diagnostic superclasses, to support triage of resting ECGs in outpatient clinics" is. The examples below indicate the range of acceptable problem classes.

Problem class	Example problem statement	Example open data sources
Binary classification	Detect pneumonia from a chest radiograph	PneumoniaMNIST; open chest X-ray collections
Multi-class classification	Assign one of seven diagnostic categories to a dermatoscopic image	HAM10000; DermaMNIST
Multi-label classification	Assign one or more diagnostic statements to a 12-lead ECG	PTB-XL
Regression	Predict length of stay from admission variables	UCI Diabetes 130-US Hospitals
Time-series prediction	Forecast a physiological signal, or predict deterioration ahead of onset	Sleep-EDF; open PhysioNet Challenge sets

Problem class	Example problem statement	Example open data sources
Image segmentation	Delineate a lesion or organ boundary	Kvasir-SEG; Medical Segmentation Decathlon
Clustering / phenotyping	Identify unsupervised patient subgroups and characterise them	NHANES; BRFSS
Anomaly detection	Flag rare or abnormal events in a physiological recording	MIT-BIH; PTB-XL
Text / NLP	Classify or summarise de-identified clinical or biomedical text	PubMed abstracts; open clinical-text corpora

4.4 Selecting your topic

By Friday, October 2, 2026 each team selects its dataset and defines its problem, and records both in the initial README of its repository. The README must name the dataset and where it comes from, state the problem you will solve and why it matters, and identify the task type, for example binary classification, multi-label classification, regression, segmentation or clustering. Submit the repository URL on myCourses by the same deadline. Teams are free to choose the same dataset or a similar problem as another team; what is assessed is your own work, and independent teams working on related problems is normal in research. Copying another team's code, text or results is not, and remains an academic offence.

5. Technical requirements

5.1 Models

You must train and compare at least two models, and three where the task allows:

1. A statistical or trivial baseline. A majority-class or mean prediction, a single-feature rule, or a simple logistic or linear regression. This establishes the floor and is not optional.
2. A classical machine learning model, for example a regularised regression, a support vector machine, a random forest, or gradient boosting.
3. Where the task and the data allow, a modern deep learning model appropriate to the modality, for example a convolutional network, a recurrent or temporal model, a transformer, or a fine-tuned pretrained backbone.

Two models are the floor and three are expected wherever the modality is a natural fit for deep learning, which includes most imaging, signal and text tasks. On a small tabular dataset a well-tuned classical model may legitimately beat a network, and demonstrating that with evidence is a better result than forcing a deep model onto data that cannot support one. Say in the manuscript why you chose the families you did. Further models, ablations or hyperparameter studies are welcome, but more models are not automatically better: a clean comparison of two or three beats a confused comparison of six.

5.2 Evaluation protocol

- **Declare your splits before you model.** Training, validation and a held-out test set must be defined at the start and reported in the manuscript. Where the data have grouping structure, such as several images per patient or several recordings per subject, splits must be made at the group level, never at the row level.
- **Tune on validation, report on test.** The test set is touched once. Reporting a test score that was used to select a model is data leakage and will be penalised heavily, in this course and in every venue you will later publish in.
- **Use metrics appropriate to your task and class balance.** For imbalanced classification, report both AUROC and AUPRC, and report sensitivity and specificity at a stated operating threshold. Accuracy alone is not acceptable for imbalanced data.
- **Quantify variability.** Report either repeated runs across several random seeds or confidence intervals on your primary metric. A single number with no uncertainty is not a result.
- **Produce one calibration or interpretability artifact**, for example a reliability curve, a feature importance or SHAP analysis, a saliency or Grad-CAM overlay, or a confusion matrix with error analysis. It belongs in the manuscript, and in your interface too if you build one.

> *The most common way student medical-AI projects fail is not weak accuracy. It is an accuracy figure that turns out to be meaningless. We grade the protocol as heavily as the numbers.*

5.3 Interface (optional, bonus available)

- Building an interface is OPTIONAL. No mark in the rubric in Section 7.2 depends on it, and a team that builds none is not disadvantaged. Teams that build one and deploy it publicly earn a bonus of 2 percentage points, described in Section 8.3. That bonus is the only credit attached to the interface.
- If you do build one, it should take an input of the kind your model consumes and return the prediction, a measure of confidence, and your interpretability output. Lightweight Python application frameworks intended for machine learning demonstrations are recommended; free hosting suitable for such applications is described on myCourses.
- Any interface you build or deploy must display, prominently and at all times, a notice that it is a course prototype for research and educational purposes, that it is not a medical device, and that it is not for clinical use. It should also handle an invalid or unexpected input without crashing.

5.4 Repository (required; public release optional, bonus available)

Every team MUST maintain a repository and MUST give the instructors and the teaching assistants read access to it, because the repository is graded. On GitHub, add @AHooshiar as a collaborator; the teaching assistants' handles are posted on myCourses. On another platform, grant the equivalent read access and confirm it in your myCourses submission. A private repository with that access granted satisfies this requirement in full. Making the repository public is OPTIONAL and earns a bonus of 3 percentage points, described in Section 8.2. Use any mainstream hosting platform, such as GitHub, GitLab or Bitbucket. Whether public or private, the repository must contain:

- A README. Its first version, committed by the selection deadline, must already name the dataset, its source and licence, and the problem you will solve. By submission it must also state how to obtain the data, how to reproduce the results, the headline results table, and the URL of the interface if you deployed one.
- All code used to obtain, prepare, train, and evaluate, as scripts or notebooks, organised so that a reader can follow the order in which they run.
- An environment specification: a requirements or environment file pinning the packages you used.
- A short model card for your best model: intended use, training data, evaluation data, metrics, known limitations, and populations on which it should not be used.
- An open-source licence (MIT or Apache 2.0 are recommended) that is compatible with the licence of your dataset.
- A contribution statement in CRediT format (see the course outline), and a Use of Generative AI statement.

Do not commit the dataset itself. Provide a download script or documented instructions. Do not commit model weights larger than about 100 MB; host them externally and link to them. Do not commit credentials or access tokens of any kind.

Commit history is part of the assessment. Each member is expected to commit under their own account, throughout the term. A repository with a single commit made on December 11 by one person will be treated as evidence about how the work was distributed, and the peer evaluation and contribution statement will be read in that light.

6. Midterm Report, due Monday, November 9, 2026 (20%)

A short technical paper defining your project and reviewing the relevant literature.

- Length: 3 pages, IEEE conference two-column format, references included. The page limit is enforced.
- Format: prepared either in Word or in LaTeX. IEEE provides official templates for both; links are on myCourses. Submit the compiled PDF and the editable source (the .docx file, or the .tex file together with the .bib file and any figures).
- References: at least 15, predominantly primary research literature, and predominantly from the last five years. Reviews and textbooks may support but not replace primary sources.

Required content: problem statement and clinical motivation; a synthesised review of prior work on this problem or closely related ones; a quantitative description of your dataset (size, classes, distributions, known limitations); your planned evaluation protocol including splits and metrics; your candidate models; and a work plan with role assignments and identified risks. A CRediT contribution statement and a Use of Generative AI statement are required, and neither counts toward the page limit.

Every reference must have been read by at least one team member. We ask about specific citations.

Midterm marking rubric

Criterion	What we look for	Weight
Problem definition and clinical significance	The problem is specific, answerable with the chosen data, and its clinical relevance is argued, not asserted	25%
Literature review	At least 15 references, predominantly primary literature from the last five years; synthesised into an argument rather than listed	35%
Data and feasibility plan	Dataset described quantitatively; splits, metrics, candidate models and compute budget stated; risks identified	25%
Format, writing and referencing	Two-column IEEE layout, page limit respected, consistent citation style, figures and tables captioned and referenced, required statements present	15%

7. Final Deliverables, due Friday, December 11, 2026 (45%)

Four items, submitted together on myCourses:

1. The manuscript, as a PDF, together with its editable source.
2. The recorded presentation, at most 7 minutes (see Section 8.1).
3. The URL of the repository.
4. The individual peer evaluation form, submitted separately by each student.

7.1 Manuscript

- Length: 6 to 8 pages, IEEE conference two-column format, references included. Papers shorter than 6 or longer than 8 pages are penalised.
- Format: Word or LaTeX, as for the midterm. Submit the PDF and the source.
- Structure: title and authors; abstract of at most 250 words; introduction and clinical motivation; related work; data; methods; results; discussion including limitations; conclusion; CRediT contribution statement; Use of Generative AI statement; references.
- The results section must include the model comparison table with uncertainty, the interpretability or calibration artifact, and at least one figure that is your own work rather than a reproduction.
- The discussion must state honestly what your model cannot do and what would be required before anything like it could be used clinically.

The midterm report may be extended into the manuscript; self-overlap between the two is expected and permitted.

7.2 Marking rubric

Criterion	What we look for	Weight
Manuscript	Scientific quality: clear question, honest results, limitations acknowledged, conclusions supported by the evidence presented	30%
Modelling rigour	Correct and declared data splits, appropriate metrics, meaningful comparison across models, evidence of no test-set leakage	35%
Recorded presentation	Clear scientific narrative within 7 minutes, results and limitations covered, all members introduced	15%
Repository and reproducibility	A competent reader can rerun the work from the repository: documented, licensed, environment specified, no data or large weights committed	15%
Contribution and disclosure	CRediT contribution statement and Use of Generative AI statement present, complete, and consistent with the commit history	5%

The mark produced by this rubric is a team mark. Each student's individual mark on this component is then adjusted by up to ± 5 percentage points on the basis of peer evaluation, capped at the component maximum, as described in the course outline. The adjustment is symmetric and is read alongside the contribution statement and the commit history.

8. Recorded presentation and public release

8.1 Recorded presentation (required)

There is no live final presentation in this course. Each team instead submits one recorded presentation, reviewed by the instructors and the teaching assistants. The requirements are below; the presentation carries 15% of the Final Deliverables mark.

Requirement	Specification
Length	Maximum 7 minutes. The recording is not watched beyond 7:00, and content after that point is not marked.
Participation	All team members must appear on camera and introduce themselves by name. The scientific content may be delivered by one member or shared among several.
Content	Structured as a scientific presentation: the problem and why it matters, the data, the models compared, the results with their limitations, and the conclusion. If you built an interface, demonstrate it on screen; if you did not, walk through your results and key code instead.

Requirement	Specification
Submission	Uploaded to myCourses by the final deadline, as a video file or as a private link. It is reviewed by the instructors and the teaching assistants only, and is not circulated to the rest of the class.
Production	A screen recording with narration is entirely sufficient. Marks follow clarity and scientific content, not production value.

8.2 Bonus: public release of the codebase (+3%)

Sharing the repository with the instructors and the teaching assistants is mandatory, because the repository is graded. Making it public is not. Teams that do publish their complete project codebase receive a bonus of 3 percentage points on the final course grade. To qualify, the repository must be publicly readable without a login by anyone with the URL, from the date of submission until at least May 1, 2027; must carry an open-source licence compatible with the licence of your dataset; and must meet in full the contents requirements of Section 5.4.

A team that prefers to keep its code private is assessed on exactly the same rubric and is not disadvantaged. Choose deliberately: a public repository is permanent and attributable to each of you by name, which is both why it is worth doing and why it should not be published before you are satisfied with it.

8.3 Bonus: publicly deployed interface (+2%)

Teams that deploy their interface so that anyone can use it from a public URL, without a login, receive a bonus of 2 percentage points on the final course grade. Free hosting suitable for machine learning demonstrations is described on myCourses. To qualify, the deployment must be live and reachable from the submission deadline until at least May 1, 2027, must run your own trained model rather than a placeholder, and must display the prototype notice required in Section 5.3.

The two bonuses are independent and may be earned together, for a maximum of 5 percentage points. Both are added to the final course grade, which remains capped at 100%.

9. Final session, Thursday, December 3, 2026

The final session is a course wrap-up with time set aside for questions about the projects, eight days before the final deadline. Bring the problems you are stuck on. Nothing is submitted or marked in this session.

10. Academic integrity, generative AI, and data ethics

- All submitted work must be your team's own. Code, figures or text taken from any external source, including public repositories, tutorials, and published papers, must be attributed at the point of use.
- Generative AI tools may be used for coding assistance, debugging, and language editing. Every deliverable must carry a Use of Generative AI statement, and every multi-author deliverable a CRediT contribution statement, in the form required by the course outline. Read that section before you submit; undisclosed use is an academic offence under the Code of Student Conduct and Disciplinary Procedures.
- You remain responsible for everything you submit. "The model produced it" is not a defence for an incorrect method, a fabricated citation, or a result you cannot explain.
- Respect the licence and the terms of use of your dataset, and cite it as its custodians request.
- No identifiable personal health information may be used, processed, or published, under any circumstances. Data that is already public, de-identified and openly licensed for educational or research use is acceptable; anything else obtained from a hospital or clinical service is not.
- If you publish your repository, anything you commit is permanent and attributable to you by name.

11. Support

- Teaching assistants are available by email for project questions; their addresses are posted on myCourses.
- A myCourses discussion forum is available for technical questions. Post there first rather than emailing: answers benefit everyone, and duplicated questions are common.
- IEEE templates for Word and LaTeX can be found here: https://www.ieee.org/conferences/publishing/templates
- Questions about the interpretation of this specification should be directed to Dr. Hooshiar.

> *This specification may be updated during the term. Any change will be announced on myCourses and in class, and the version date in the footer will change.*